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Structure Analysis of Salivary Amylase

by

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1.Introduction

Amylase is a digestive enzyme that hydrolyzes the glycosidic bonds in starch molecules to cleavage carbohydrates into simple sugars, making the nutrients easier to be absorbed by the owner. There are three types of amylase enzymes (α -, β -, and γ -amylase), each of them catalysis on different parts of the carbohydrate molecule[1, 2]. Human salivary amylase, produced by salivary glands, is a type of α -amylase that breaks down the starch, maintaining body's energy supply (Fig.1). It's a key component of human salivary and plays a crucial role in human life support and metabolism[3]. Biomarkers remain salivary amylase's major application. The activity of the enzyme is frequently utilized as an indicator for stress processes in various nervous systems and the associated metabolic activities, becoming an important element in relevant scientific research⁴. Moreover, it may have potential clinical diagnostic capacity[4]. Research has found that it could be a signal of diseases such as obesity, Fibromyalgia (FM), cardiovascular (CV) diseases, and salivary disease[5-7].

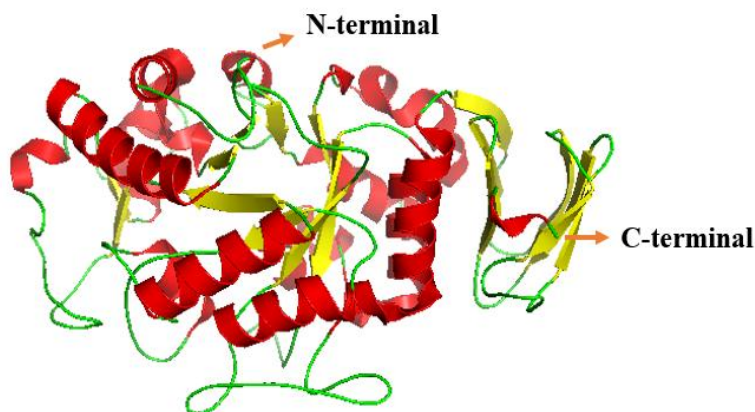


Figure.1 The three-dimensional (3D) model of human salivary amylase secondary structure.

2.Materials and methods

This study analyzed the secondary structure of human salivary α -amylase. The protein structure was obtained during the research of substrate-binding site amino acids Trp58 by inducing residues alternation[8].PDB coordinates of amylase was first got from PDB database with PDB code 1jxj.Then, the enzyme statistics was uploaded to PDBsum for

further analysis by both PDB codes and PDB structure files. After comparison, results from PDB code were chosen to analysis due to more comprehensive information. The protein quality assessment was conducted by the program of PDBsum. Ramachandran plot was produced to help evaluate the protein structure's quality, improving its accuracy. The 3D models of the amylase and the active sites visualization were performed in PyMol program. Pertinent scientific literature in this study was primarily from the PubMed database.

3. Protein secondary structure analysis

3.1 Protein structure composition

The salivary amylase had 496 amino acids in total, which formed 8 sheets, 5 beta alpha beta units, 6 beta hairpins, 3 beta bulges, 24 strands, 21 helices, 11 helix-helix interac, 51 beta turns, 5 gamma turns, and 5 disulphides (Fig.2). Two cofactors were also present, which were metal ions CL (chloride ion) and CA (calcium ion). The detailed information of the structure was listed in Table.1.

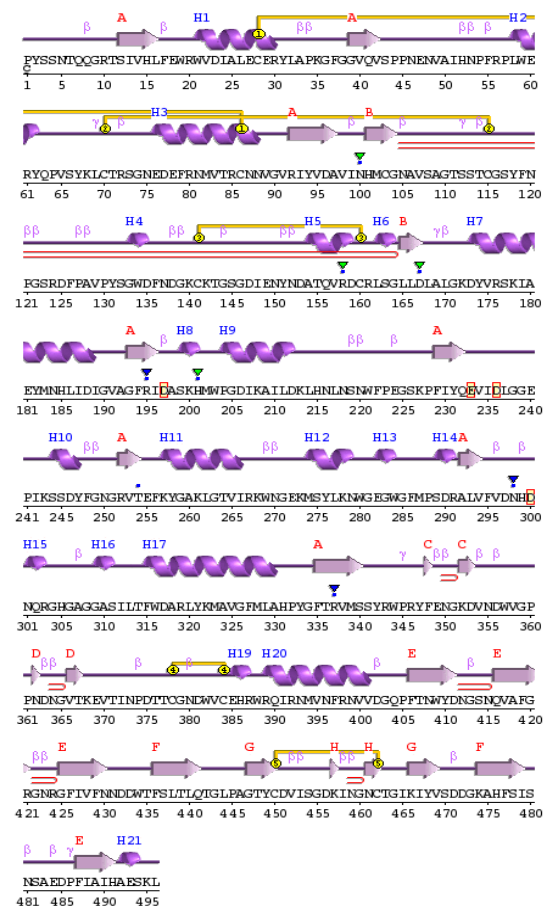


Figure.2 The secondary structure of human salivary amylase (1jxj). Helix represented the α -helix structure and the arrow represented the β -sheet structure. Motifs (β -turns, γ -turns, and β -hairpins), disulfide bonds, catalytic residues, and metal contacts were also annotated.

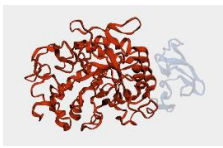
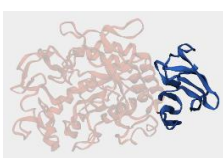
CATH Domain	Description	α -helix position	β -sheets position
1 (residue:2-403) 	Mainly contains α - β Barrel structure	20-31, 57-63, 75-89, 132-135, 153-160, 172-190, 198-202, 203-213, 243-248, 256-267, 273-280, 281-285, 288-290, 300-304, 308-312, 314-317, 318-331, 384-388, 389-402	11-17, 38-42, 92-97, 100-105, 164-168, 192-197, 229-233, 252-255, 291-294, 335-340, 347-349, 352-254, 361-363, 366-368,
2 (residue:404-496) 	Mainly contains β -sandwich structure	493-495	405-412, 415-422, 424-431, 436-442, 447-451, 456-458, 461-463, 465-469, 473-479, 486-492

Table.1 Domains presented in the amylase 1jxj, which had 20 α -helix and 24 β -sheets.

3.2 Ramachandran plot (structure quality assessment)

Ramachandran plot described the energetically permissible regions of backbone dihedral angles ϕ (phi) and ψ (psi) (peptide bond orientation) for amino acids in protein structures. It was frequently used to evaluate and appraise the quality of protein structures. In the assessment, G factor scores estimated the unusual level of the protein structure.

The statistics and Ramachandran plot of amylase 1jxj were presented in Table.2 and Figure.3. It could be concluded that most of the residues were in energetically allowed regions. Meanwhile, the G factor scores of 1jxj was significantly higher than the unusual scores standards, confirming the validation of the protein structure.

Ramachandran plot statistics	
End-residues (excl. Gly and Pro): 2 Glycine residues: 51 Proline residues: 22	Non-glycine and non-proline residues: 420
Regions	Percentage (non-glycine and non-proline residues)
Most favoured regions [A,B,L]	88.3%
Additional allowed regions [a,b,l,p]	11.4%
Generously allowed regions [~a,~b,~l,~p]	0.2%
Disallowed regions [XX]	0%
G factor overall average scores	Unusual standard scores
0.04	-0.5(unusual)/-1.0 (highly unusual)

Table.2 The statistics of amylase 1jxj Ramachandran plot.

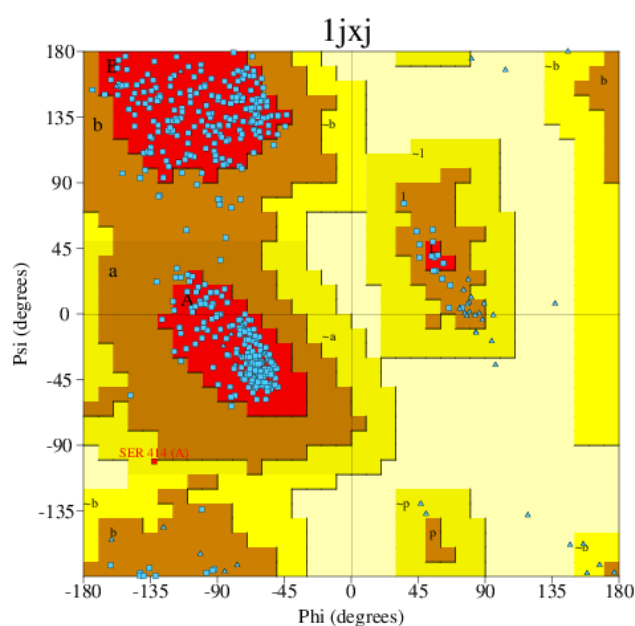


Figure.3 The Ramachandran plot of amylase 1jxj. The majority of peptide angle scores were spread within permitted regions.

3.3 Active sites and cofactors

The protein had 3 active sites, which were listed in Table.3 with their functional annotations and 3D visualization. In particular, all of the sites were found to be in the same location, which was in the middle and directly in front of a β -barrel. Two cofactors CA (calcium ion) and CL (chloride ion) were also included in the protein.

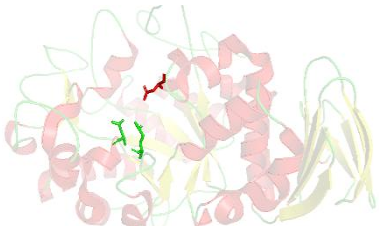
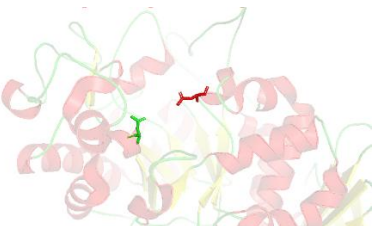
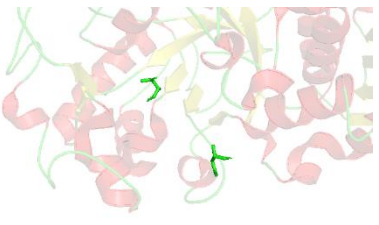
Active site	Function	Visualization
Site I (Asp197 - Asp300 - Glu233)	Endohydrolysis of (1->4)- α -D-glucosidic linkages in polysaccharides containing three or more (1->4)- α -linked D-glucose units.	
Site II (Asp197 - Glu233)		
Site III (Asp197 - Asp236)	Simultaneously produces isomaltulose (6-O-alpha-D-glucopyranosyl-D- fructose) and smaller amounts of trehalulose (1-O-alpha-D-glucopyranosyl-beta-D-fructose) from sucrose.	

Table.3 Annotation of active sites in salivary amylase 1jxj.

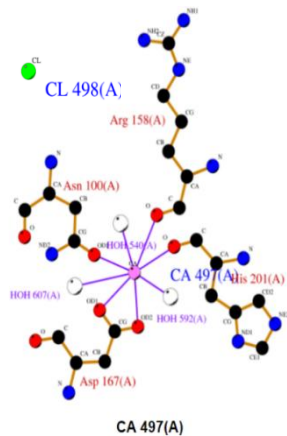


Figure.4 Metal ions CL (upper) and CA (lower) in the amylase, acting as enzyme cofactors. CA was responsible for the main cofactor function with ligand binding to 4 amino acids of the amylase while CL was allosteric regulator bond closely to active sites.

4. Discussion and conclusion

Human salivary amylase is a vital enzyme for breaking down ingested starch and is involved in several crucial metabolic processes. In scientific study, the fluctuation in enzyme concentration is frequently utilized as a biomarker[4]. Enhancing enzyme-specific antibodies is important for better enzyme detection. Therefore, the study of its secondary structure is crucial for advancement in related fields of science.

This study mainly focused on the analysis of the secondary structure of human salivary α -amylase. From the structure arrangement results (Fig.2), it could be concluded that the enzyme had a unique structure shaped by various forms of α -helixes and β -sheets, and diverse non-covalent interactions, which provided a base for the substrates binding and cleavage reactions. Loops were also vital in this enzyme. Crystallographic research has found that the flexible, glycine-rich loop ³⁰⁴GHGAGGA³¹⁰ presented in the amylase might facilitate the release of products from the active site[9].

The enzyme had 3 catalytic sites that consisted of aspartic acid (Asp197, Asp236, Asp300) and glutamic acid (Glu233) (Table.3). Site I and Site II were responsible for hydrolysis of the (1- $\rightarrow$ 4)- α -D-glucosidic linkages in polysaccharides while Site III functioned isomerization of the sucrose. Asp and Glu were common amino acids that found in the active sites of enzymes. Both of them contained carboxylic acid functional groups in their side chains and negatively charged under physiological pH (salivary environment), donating protons. As a result of this property, the enzyme would interact with other molecules. The special location of the catalytic residues in front of the β -barrel may serve as a conduit for the transportation of the sugars to the active sites. Moreover, the enzyme required 2 necessary metal cofactors (CA and CL) for its activity (Fig.4). Calcium ion was necessary for the activation of α -amylase with binding of Asn100, Arg158, Asp167, His201. Chloride ion ligand bound near the catalytic sites and was thought to work as an allosteric activator[10].

Previous study found that human pancreatic α -amylase and salivary α -amylase had comparable but distinct cleavage patterns. This difference may be caused by the conformation variance. Compared to human pancreatic α -amylase, the salivary α -amylase had 3 extra polar groups and 3 charged side chains due to the discrepancy of 15 amino acids. Two of these (Thr163 to serine and Leu196 to isoleucine), in the active sites area, might affect the function of enzymes[10, 11].

The study generally used PDB database for protein information, PDBsum database for structure analysis, PyMol for 3D model construction, and PubMed database for relevant articles. PDB and PDBsum are mainstream databases for scientific protein research, while PubMed is an authority MEDLINE database for biomedical study. Therefore, the protein data and relative references were reliable and valid. The quality of the enzyme structure was also evaluated by Ramachandran plot. The plot results proved the validity of the study statistics, again. Furthermore, 3D models were created to more directly expose the enzyme conformation and active regions, offering a thorough examination of protein characteristics.

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